

Interview Q&A: tensorflow

1. What is TensorFlow?

TensorFlow is an open-source software library for numerical computation, particularly well-suited for large-scale machine learning tasks.

2. What are the main differences between static graph execution and eager execution in TensorFlow?

Static graph execution is a traditional way of executing TensorFlow graphs, where the computation graph is built beforehand and executed in a single pass. Eager execution, on the other hand, executes operations immediately as they are defined.

3. What is the purpose of the `tf.data` API in TensorFlow?

The `tf.data` API provides a flexible way to load, transform, and batch data for training and evaluation.

4. What is the difference between `tf.Variable` and `tf.get_variable()`?

`tf.Variable` is a simple variable that can be used to store values, while `tf.get_variable()` is used for shared variables across multiple scopes.

5. What is the purpose of the `tf.keras` API in TensorFlow?

The `tf.keras` API provides a high-level interface for building and training neural networks, making it easier to implement common deep learning workflows.

6. What is the purpose of the `tf.Session` object in TensorFlow?

The `tf.Session` object is used to run TensorFlow operations on a device, such as a GPU or CPU.

7. What is the purpose of the `tf.keras.layers` module in TensorFlow?

The `tf.keras.layers` module provides a high-level API for building neural networks, including pre-built layers such as convolutional and recurrent layers.

8. What is the difference between `tf.function` and `@tf.function` decorators in TensorFlow?

The `tf.function` decorator is used to convert a Python function into a TensorFlow graph, while the `@tf.function` decorator is a shortcut for creating a decorated function that can be converted into a graph.

9. What is the purpose of the `tf.config` API in TensorFlow?

The `tf.config` API provides a way to configure various aspects of the TensorFlow runtime, such as setting up GPUs or TPUs for computation.

10. What is the purpose of the `tf.keras.Model` class in TensorFlow?

The `tf.keras.Model` class is a high-level API for building neural networks, providing an easy-to-use interface for defining and training models.

11. What is the purpose of the `tf.keras.optimizers` module in TensorFlow?

The `tf.keras.optimizers` module provides a set of pre-built optimizers that can be used for training machine learning models.